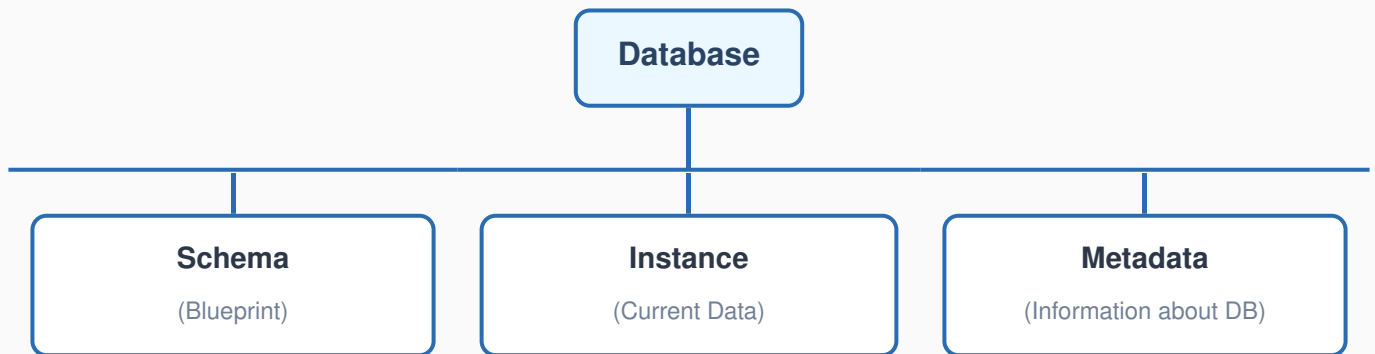


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Database Core Concepts

Every database has three important things:



Think of them as:

Schema	→	Design
Instance	→	Current State
Metadata	→	Description of the Design .

A **database schema** is the logical structure or blueprint of a database that defines how data is organized, including tables, attributes, relationships, constraints, and other database objects.

A schema defines:

- Tables
- Columns
- Data types
- Constraints
- Relationships
- Views
- Indexes .

suppose we execute:

```
ALTER TABLE Students  
ADD Email VARCHAR(100);
```

The schema has changed.

Changing schemas requires:

- Planning
- Testing
- Migrations
- Application updates

Therefore,
schema changes happen much less frequently than data changes.
This is why we say the schema is *relatively static*, *not absolutely static*.

A **database instance** is the collection of data stored in the database at a particular point in time.

Every insert, update, or delete creates a new database instance.

Metadata & Data Dictionary

Metadata is data that describes other data.

Simple.

It doesn't store your actual business data.

Instead,

it stores information about the database.

A **data dictionary** is a repository that stores metadata about the database.

It contains:

- Table definitions
- Column definitions
- Constraints
- Keys
- Data types
- Relationships
- Permissions
- Views
- Index definitions.

System Catalogue

System catalogue : Many confuse it with the data dictionary.

Definition

The **system catalog** is the DBMS-maintained collection of system tables that stores metadata about all database objects.

The system catalog stores the data dictionary (metadata).

Where Is Metadata Stored?

Metadata is stored by the DBMS inside the system catalog (or catalog tables), which together form the data dictionary.

The system catalog is the DBMS-maintained collection of system tables that stores metadata. It is the physical implementation through which the DBMS maintains the data dictionary.